

Longitudinal Inference Without Longitudinal Data

Estimating Mean *SGPc* Without Linked Data using Copula-Induced Stochastic Growth Regimes

Synthetic low-growth example: True mean *SGPc* = 39.0, baseline *t*-copula ($\rho = 0.72$, $df = 8$)

$W_1(F_{V,S}^{obs}, F_H)$: Uniform = 0.0915 → Best-fit = 0.0019 (97.9% reduction)

A. Convert scores to reference quantiles

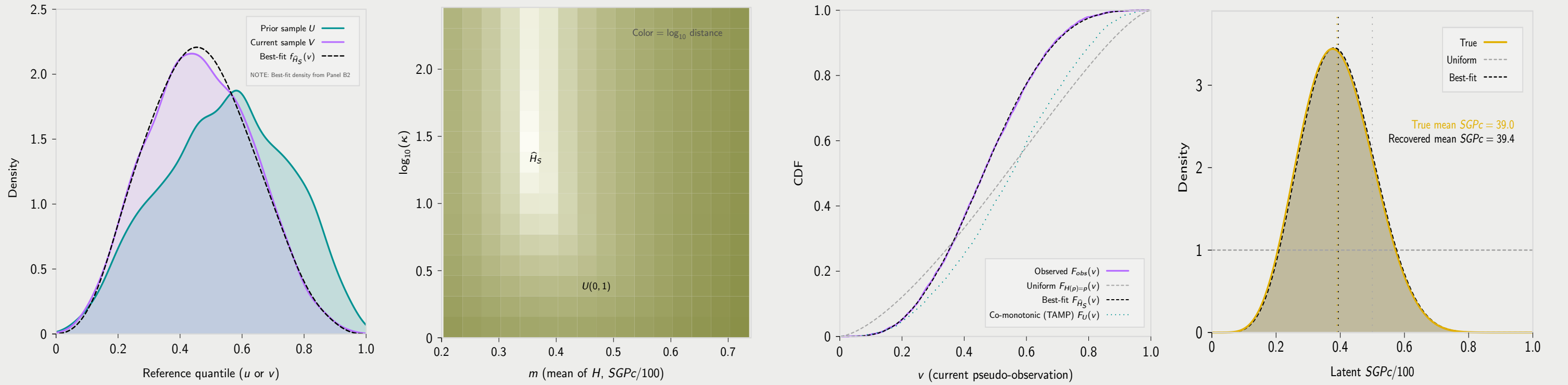
Observed, non-linked marginals: U (prior) and V (current)

B1 & B2. Estimate subgroup growth regime $\hat{H}_S$ (copula + regime $\Rightarrow$ induced marginal)

Choose $H \in \mathcal{H}_{\text{Beta}}$ to minimize $W_1(F_{V,S}^{obs}, F_H)$, where $F_H(v) = \mathbb{E}_U [H(F_0(v | U))]$

C. Recover inferred latent *SGPc* density $P_S \sim \hat{H}_S$

Report density/summary: $\text{meanSGPc} = 100\hat{m}$



We observe two independent cross-sectional, non-longitudinal samples of the same subgroup S (e.g., a TIMSS country or a NAEP state): prior scores X at time t_1 and comparison scores Y at time t_2 . The design does not require that t_2 represent a later measurement of the same cohort: (t_1, t_2) may index different grades within the same year (e.g., Grade 4 vs. Grade 8), or the same grade measured in different years (often interpreted as $t_2 = t_1 + 4$ under a same-cohort, four-year-later convention).

We map scores to reference-quantiles (pseudo-observations) using fixed reference marginal CDFs:

1. Prior reference quantiles $U = F_X^{\text{ef}}(x_i)$ and
2. Current reference quantiles $V = F_Y^{\text{ef}}(y_j)$.

Crucially, there is *no student-level linkage* between these samples—we cannot match any u_i to its corresponding v_j .

Panel A shows the two *univariate* marginals plotted on the same reference-quantile axis (“ u or v ” in $[0, 1]$). This is the information available in TIMSS/NAEP-style designs when cohorts are sampled independently (with survey weights, when applicable).¹

After the reference-quantile transform, U and V summarize the observable evidence. All subsequent inference proceeds from these marginals plus an externally supplied copula dependence structure.

The mathematical bridge connecting the observed, unlinked marginals (Panel A) and the recovered latent *SGPc* regime (Panel C) has two ingredients:

1. A baseline copula C_0 providing a dependence template from prior reference-quantiles U to current reference-quantiles V .²
2. A stochastic growth regime H that governs how subgroup S allocates conditional percentiles within that template.^{3, 4}

Together, C_0 and H define a *forward map* from a candidate regime to a *predicted* current-grade marginal CDF. Panel B2 visualizes this map via the induced $F_H(v)$, and Panel B1 *inverts* it by choosing the regime whose F_H best matches the observed current marginal $F_{V,S}^{obs}$.

Let C_0 be a baseline copula. C_0 induces the conditional CDF (transition kernel)

$$F_0(v | u) = \Pr(V \leq v | U = u) = \frac{\partial}{\partial u} C_0(u, v).$$

Its inverse in v (the conditional quantile kernel) is

$$Q_0(p | u) = F_0^{-1}(p | u).$$

Intuitively, for each starting quantile u , the kernel $F_0(\cdot | u)$ describes the distribution of possible ending quantiles V under the baseline dependence structure C_0 .

Let $P \in [0, 1]$ be a random variable (a latent *SGPc* on the 0–1 scale) and let p denote a realization in $[0, 1]$. A *growth regime* H is the distribution of P , represented by its CDF

$$H(p) := \Pr(P \leq p), \quad p \in [0, 1].$$

Given $U = u$ and a draw $P = p$, the model generates an outcome percentile by:⁵

$$V = Q_0(P | U = u).$$

The uniform regime $H(p) = p$ (i.e., $P \sim \text{Uniform}(0, 1)$) corresponds to occupying the baseline kernel with no tilt. Departures from $H(p) = p$ tilt the growth regime mass toward lower or higher conditional percentiles relative to the baseline regime.

Although we do not observe linked pairs (u, v) in TIMSS- or NAEP-style settings, the model implies a forward map from the observed prior marginal F_U to the predicted current marginal F_H . For any fixed u ,

$$\begin{aligned} \Pr(V \leq v | U = u) &= \Pr(Q_0(P | u) \leq v) \\ &= \Pr(P \leq F_0(v | u)) \\ &= H(F_0(v | u)). \end{aligned}$$

Therefore, by the law of total probability, the subgroup’s predicted current-grade marginal CDF is

$$\begin{aligned} F_H(v) &= \Pr(V \leq v) \\ &= \mathbb{E}_U [H(F_0(v | U))] \\ &= \int_0^1 H(F_0(v | u)) dF_U(u) \\ &\approx \frac{\sum_{i=1}^n w_i H(F_0(v | u_i))}{\sum_{i=1}^n w_i}, \end{aligned}$$

where $\{u_i, w_i\}$ are the observed prior pseudo-observations and survey weights for the subgroup (i.e., a weighted empirical approximation to F_U).

We estimate the subgroup’s growth regime by searching over an admissible class $\mathcal{H}$ of candidate CDFs on $[0, 1]$. Panel B1 shows the heat map of $\mathcal{H}$ over the *Beta regime family*

$$\mathcal{H}_{\text{Beta}} = \{H_{m,\kappa}(\cdot) : P \sim \text{Beta}(\kappa m, \kappa(1 - m)), 0 < m < 1, \kappa > 0\},$$

where $m = \mathbb{E}[P]$ controls the regime center (mean *SGPc* on the 0–1 scale) and κ controls concentration (dispersion of latent *SGPc*).⁶

For each candidate $H \in \mathcal{H}$, Panel B2 provides the induced *predicted* current-grade marginal

$$F_H(v) = \mathbb{E}_U [H(F_0(v | U))].$$

The growth regime estimate for subgroup S is defined by the minimum-distance problem

$$\hat{H}_S = \underset{H \in \mathcal{H}_{\text{Beta}}}{\operatorname{argmin}} W_1(F_{V,S}^{obs}, F_H),$$

where

$$W_1(F, G) = \int_0^1 |F(v) - G(v)| dv$$

is the Wasserstein-1 (earth mover’s) distance between the observed and predicted current-grade marginal CDFs.⁷

The heat map displays $\log_{10}$ Wasserstein-1 distance over $(m, \log_{10} \kappa)$: lighter regions indicate candidate regimes whose induced F_H more closely matches $F_{V,S}^{obs}$. The marked optimum corresponds to $\hat{H}_S$ (equivalently, $(\hat{m}, \hat{\kappa})$). For reference, the uniform regime $H(p) = p$ occurs at $m = 0.5$ and $\kappa = 2$ (i.e., $\text{Beta}(1, 1)$), indicated by the $U(0, 1)$ marker.

The baseline copula, C_0 , combined with the best fit growth regime estimate, $\hat{H}_S$, induces the Wasserstein-1 minimized CDF, $F_{\hat{H}_S}(v)$, and its density, $f_{\hat{H}_S}(v)$, which are shown in panel B2 and A, respectively, with a dashed-black line.

The output of Panels B1–B2 is a subgroup-specific *estimated growth regime* $\hat{H}_S$, where $\hat{H}_S$ is a cumulative distribution function (CDF) on $[0, 1]$ for a latent conditional-percentile random variable P_S (*SGPc* on the 0–1 scale):

$$\hat{H}_S(p) = \Pr(P_S \leq p), \quad 0 \leq p \leq 1.$$

In words, $\hat{H}_S$ describes how subgroup S allocates conditional ranks *within* the baseline dependence template $F_0(\cdot | u)$.

Panel C displays the *distribution of* P_S (typically via its density $f_{\hat{H}_S}(p) = \hat{H}'_S(p)$ when $\hat{H}_S$ is smooth), not the CDF itself. Thus the “Best-fit” curve is the density corresponding to the fitted CDF (i.e., $P_S \sim \hat{H}_S$). The dashed grey “Uniform” reference corresponds to the uniform regime $H(p) = p$ (equivalently $P \sim \text{Unif}(0, 1)$), which places no tilt on conditional percentiles relative to the baseline kernel.

Because we restrict the candidate regime class $\mathcal{H}$ to the Beta family in Panel B1, the fitted regime implies $P_S \sim \text{Beta}(\kappa m, \kappa(1 - m))$: the mean $m = \mathbb{E}[P_S]$ sets the regime center (reported as mean *SGPc* = $100m$), and κ controls concentration of latent *SGPc*.

The vertical reference lines mark mean *SGPc* under the uniform regime (50), the generating “True” regime (39.0), and the recovered best-fit regime (39.4). In applications, the primary outputs are $\hat{H}_S$ (with uncertainty bands) and summary functionals such as mean/median *SGPc* and tail/bucket probabilities.

¹ Panel A also shows the density, $f_{\hat{H}_S}(v)$, the density induced from the baseline copula and fitted growth regime from panels B1 and B2, superimposed on the observed density, $f_{V,S}^{obs}(v)$.

² The baseline copula used for inference is selected from canonical STEP 1 sensitivity analyses over 966 longitudinal conditions across four datasets: the *t*-copula is best by AIC in 63.6% of conditions (Frank: 30.7%). In the cross-stratified canonical table below (content area $\times$ year span), the recommended family is *t* in every cell; entries report (ρ, df) medians.

Content area	1-year	2-year	3-year	4-year
MATHEMATICS	(0.85, 25.26)	(0.81, 27.40)	(0.78, 28.38)	(0.76, 27.34)
READING	(0.84, 28.33)	(0.80, 29.34)	(0.77, 26.63)	(0.76, 28.22)
WRITING	(0.85, 23.72)	(0.82, 27.58)	(0.80, 27.22)	(0.77, 23.86)
ELA	(0.84, 41.95)	(0.79, 46.41)	(0.76, 51.57)	(0.72, 42.04)

³ Other regime families considered include the truncated exponential family (the maximum-entropy distribution on a finite interval with a fixed mean) and the truncated uniform family. If acceptable accuracy and precision are already achieved under the Beta family, additional growth-regime optimization and fine-tuning should only improve the recovered fit.

⁴ Here, U denotes the subgroup’s prior-grade reference quantile (pseudo-observation), and C_0 is the canonical baseline copula selected from STEP 1 by content area and year span. The fitted growth regime $\hat{H}_S$ (distribution of latent $P = \text{SGPc}/100$) and C_0 define a forward map

$$(U, C_0, \hat{H}_S) \mapsto F_{\hat{H}_S}(v) = \mathbb{E}_U [\hat{H}_S(F_0(v | U))],$$

with corresponding induced density $f_{\hat{H}_S}(v)$. The latent *SGPc* distribution itself is given by $\hat{H}_S$ (density $\hat{h}_S(p) = d\hat{H}_S/dp$), which is estimated by matching $F_{\hat{H}_S}$ to the observed current marginal.

⁵ We assume $P \perp U$. That is, p is drawn independently of starting point within the subgroup; meaning there is no correlation between the starting point u and the growth regime. This assumption is investigated more deeply in the full paper.

⁶ The Beta family, $\text{Beta}(\alpha, \beta)$, is reparameterized via $\alpha = \kappa m$ and $\beta = \kappa(1 - m)$, so that $m = \mathbb{E}[P] = \alpha/(\alpha + \beta)$ (the regime center, mean *SGPc* on the 0–1 scale) and $\kappa = \alpha + \beta$ (the concentration parameter).

⁷ Alternative minimum-distance criteria (e.g., Cramer-von Mises, $\int_0^1 \{F(v) - G(v)\}^2 dv$) are also valid for regime fitting. If acceptable accuracy and precision are already achieved under Wasserstein-1, then additional growth-regime optimization and fine-tuning should only improve the recovered fit.